

Sneha Uppu

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SUMMARY

Business Analyst and Data Analyst with 3 years of experience across healthcare technology, management consulting, and nonprofit sectors, delivering end-to-end data pipelines, predictive models, and self-service analytics platforms that inform C-suite decisions. Expert in **Python**, **SQL**, **Tableau**, and **Power BI**, with hands-on experience in **machine learning**, **NLP**, A/B testing, statistical analysis, ETL/ELT workflows, Snowflake, Apache Spark, and **GenAI/LLMs**. Track record of cross-functional stakeholder management, data governance, cohort analysis, and converting raw operational data into revenue-generating insights at scale.

EXPERIENCE

Thaddeus Resource Center

Data Science Intern

Arlington, VA, USA

Nov 2025 – Present

- Engineered an end-to-end machine learning (ML) predictive analytics pipeline in **Python** and **scikit-learn** processing 50,000+ beneficiary records, directly informing \$120K in targeted resource allocation decisions for senior program leadership.
- Developed interactive **Tableau dashboards** with 12 program KPIs across 8 service categories, enabling self-service analytics for nonprofit directors and cutting weekly reporting time from 6 hours to 45 minutes.
- Constructed **PostgreSQL** ETL pipelines consolidating 4 legacy intake systems into a unified data warehouse, reducing per-analyst data retrieval time by 3 hours per ad-hoc analysis cycle.
- Trained a **Random Forest** classification model to surface at-risk participants via feature engineering and cross-validation, flagging 340 individuals for early intervention with a 0.89 precision score on held-out test data.
- Automated 24 monthly compliance reports using **Python** (pandas, openpyxl), applying **LLM-assisted** document analysis to streamline reporting workflows, eliminate 8 hours of manual data entry per cycle, and standardize deliverables across 6 regulatory submissions annually.
- Conducted exploratory data analysis (EDA) and cohort analysis on program retention trends, combining predictive modeling and **AI-driven pattern analysis** to present recommendations to 3 executive directors that redirected \$40K in quarterly program spending.

PricewaterhouseCoopers (PwC)

Salesforce Data Analyst Intern

India

Feb 2024 – Aug 2024

- Migrated 85,000+ client records into Salesforce CRM using **Python-based** data cleansing, transformation, and validation scripts, improving the internal data quality index from 71 to 94 across 3 healthcare practice areas and resolving 12,000 duplicate records.
- Designed **A/B testing** analyses across 6 healthcare client communication workflows, applying statistical hypothesis testing and exploratory analysis to evaluate messaging variants and identify approaches associated with 1,200 incremental client engagements per quarter.
- Synthesized weekly pipeline and funnel analytics for 40+ enterprise accounts using **SQL-sourced** Salesforce data and natural-language querying to identify 3 at-risk healthcare client engagements totaling \$2.4M ahead of renewals.
- Built a **Power BI dashboard** integrating data across 5 practice areas, incorporating GenAI-assisted analytical workflows for automated data exploration and insight generation, reducing executive meeting preparation from 4 hours to 30 minutes per weekly review cycle.
- Documented 22 **functional business requirements (FRDs)** for a CRM workflow redesign through data analysis, gap analysis, and process mapping, capturing requirements for AI/ML, LLM, and RAG-enabled analytical workflows and supporting system rollout 3 weeks ahead of schedule.

Optum

Business Analyst/Data Analyst

India

Aug 2022 – Jan 2024

- Spearheaded an NLP-based insurance claims categorization model using **Python** and **NLTK**, processing 200,000+ monthly claims and eliminating 1,800 hours of annual manual review labor.
- Architected a healthcare encounter data pipeline using **Apache Spark** and **SQL Server** with automated data quality validation, cutting ETL processing latency from 6 hours to 45 minutes for 4 downstream **ML** engineering teams.
- Produced 30+ Business Requirement Documents (BRDs) and data flow diagrams across 5 **AI/ML** product initiatives, aligning 12 cross-functional stakeholders across clinical, engineering, and product domains via Agile/Scrum ceremonies.

- Validated **ML model** outputs against clinical benchmarks using statistical analysis (SciPy, statsmodels) and regression analysis, ensuring compliance with 4 CMS regulatory mandates across 2 consecutive annual audit cycles.
- Launched a self-service analytics portal in **Tableau Server** for 80+ business users, converting 14 recurring ad-hoc data requests into standardized interactive business intelligence dashboards and reducing turnaround from 2 days to 4 hours.
- Established 18 enterprise data governance and data quality rules in the governance framework, strengthening upstream reliability for 6 production **ML pipelines** serving 3 million+ annual patient records.

PROJECTS

CleanCare | GitHub

- Built a **PostgreSQL** healthcare data platform integrating **CMS** datasets across nursing homes, hospitals, and ambulatory surgical centers, exposing facility quality patterns through a 19-tool MCP server, AI chat assistant, and web dashboard with 128 automated CI tests.

Academic Paper Recommendation System | GitHub

- Engineered a hybrid research-paper recommendation system combining **BERT**, **Node2Vec**, and BM25 across 23.4 million papers and 278 million citation edges, using product quantization to reduce the FAISS index by 97%.

U.S. Airline On-Time Performance Pipeline | GitHub

- Constructed an **ETL pipeline** processing 50 million flight records across 200+ airlines into **PostgreSQL**, connecting Streamlit and Tableau dashboards that filter departure, destination, and date inputs for airline performance recommendations.

FAERS Drug Symptom Safety Analysis | GitHub

- Standardized 1M+ drug names across 18 million FDA adverse-event records from disparate source files, powering a **FastAPI** and **DuckDB** backend for querying drug-symptom co-occurrence and dosage-related safety patterns.

EDUCATION

The George Washington University <i>Master of Science, Computer Science</i>	Washington, DC Aug 2024 – May 2026
Andhra University <i>Bachelor of Engineering, Computer Science and Engineering</i>	India Aug 2020 – Apr 2024

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, DAX, Bash

ML & AI Libraries: pandas, NumPy, scikit-learn, XGBoost, TensorFlow, PyTorch, NLTK, LangChain, SciPy, BERT, Node2Vec, statsmodels, Matplotlib, Seaborn, Plotly

BI & Visualization: Tableau, Tableau Server, Power BI, Looker, Streamlit, Google Data Studio, Metabase, Advanced Excel (Power Query)

Data Engineering & Warehousing: Apache Spark, Apache Airflow, Apache Kafka, dbt (data build tool), Snowflake, Databricks, Amazon Redshift, Google BigQuery, Azure Synapse, ETL/ELT Pipelines, Data Modeling

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis, FAISS (Vector Database), Pinecone

Cloud & MLOps: Amazon Web Services (AWS), AWS SageMaker, Microsoft Azure, Google Cloud Platform (GCP), Docker, Git, GitHub, CI/CD, Jupyter Notebook

AI/ML Techniques: Machine Learning (ML), Natural Language Processing (NLP), Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Predictive Modeling, A/B Testing, Statistical Analysis, Causal Inference, Cohort Analysis, Funnel Analysis, Feature Engineering, MLOps, Information Retrieval, Recommendation Systems

Business & Methodology: Salesforce CRM, Requirements Gathering, Business Requirements Documents (BRDs), Gap Analysis, Process Mapping, UAT, Data Governance, Data Quality, Agile, Scrum, JIRA, Confluence, Cross-Functional Stakeholder Management